MD Jawadul Hasan

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Profile

Software engineer with three years of experience building data driven enterprise systems, backend services, and AI applications. Passionate about developing scalable, intelligent systems. Currently exploring how to make AI agent outputs more trustworthy by tracing answers to source evidence and distinguishing computed results from model-generated inferences.

Education

Åbo Akademi University, Turku, Finland August 2026 – Present

M.S. in Computer Science, Erasmus Mundus Joint Master's Programme (EDISS)

- Research topic (in collaboration with CNRS, France): Building a skepticism layer for LLM-based agentic systems that shows where each answer comes from, separates results computed from source data from model-generated inferences, and surfaces supporting evidence, so users can spot hallucinated claims and decide when to trust the system.
- Relevant coursework: Data Science, Cloud Computing, Artificial Intelligence, GPU Programming

American International University - Bangladesh September 2023

BSc. in Computer Science and Engineering (CSE) *Cumulative GPA: 3.99/4.00*

- Relevant coursework: Algorithms & Data Structures, OOP, Database, Networks, Web Technologies, Data Mining, Advanced Programming with Java & .NET, Software Quality & Testing, Artificial Intelligence

Skills

Core Competencies: Problem Solving, Algorithms & Data Structures, OOP, Agile, Scrum

Programming: C#, C++, Python, SQL, Java, JavaScript, TypeScript

Frameworks & Libraries: .NET, xUnit, Angular, React, Nest Js, Next Js

AI & ML: RAG, Vector Search, Embedding Models, TensorFlow, PyTorch

Database: MSSQL, MongoDB, PostgreSQL

DevOps & Cloud: Git, CI/CD, Jenkins, Docker, Azure, AWS

Architecture & Design: Clean Architecture, Microservice, Vertical Slice, MVC, REST, CQRS, SOLID, DRY, YAGNI

Professional Experience

Software Engineer March 2023 – July 2026

Brain Station 23 PLC, Dhaka, Bangladesh

- Led a team of **6** engineers delivering software solutions for large-scale enterprise clients across government services.
- Built an AI Agent to automate insurance claim processing, achieving **40%** faster submission times.
- Developed a conversational AI assistant that personalizes customer onboarding by inferring preferences from natural language, reducing manual intake effort significantly.
- Automated approval workflows supporting **10,000+** daily approvals, boosting operational efficiency through concurrency control and duplicate prevention.
- Built an integrated wallet and bank API system processing over **$3 million** in monthly payments and streamlining automated fee collection.
- Migrated **15** years of relational data with full schema mapping to improve report accuracy and reliability by **70%.**
- Designed and optimized data reporting solutions over tables of more than **50 million** records each.
- Architected a polyglot persistence strategy using MongoDB and SQL Server, reducing query latency from **12s to 500ms.**
- Migrated a legacy application to the latest .NET with comprehensive unit tests, reducing bugs by **30%**.
- Automated CI/CD pipelines with Jenkins and integrated OAuth authentication for secure third-party API connectivity.

Research Assistant (Part-time) December 2024 – August 2026

ELITE Research Lab, NY, USA (Remote)

- Pursuing applied research in machine learning and AI, focusing on generative models, predictive modeling, and computer vision.
- Developing AI-driven systems and tools, including context-aware conversational agents for real-world applications.
- Mentoring undergraduate students in LLM, AI, and machine learning research, and supporting the development of research tools and AI applications.

Projects

GradAgent.ai | *AI-Powered Graduate Admissions Assistant*

- Built an ASP.NET Core backend for an admissions chatbot, grounding LLM answers in MongoDB vector search over university and professor data.
- Scraped **20K+** professor and **1,000+** university profiles across different websites, normalizing and deduplicating the data into a structured MongoDB knowledge base for vector search.
- Implemented dynamic prompt construction and context injection to optimize LLM responses based on retrieved user profiles, university data, and application requirements.
- Developed a tool-calling LLM agent that uses RAG over university admission statistics and faculty research data to generate personalized application strategies from each student's academic profile.
- Engineered a sliding window mechanism for token allocation and rate limiting to control LLM costs across APIs.

Imbalance Price Forecasting in Electricity Markets | *Hackathon project for Turku Energia*

- Cut forecast error by **36%** and interval score by **51%** vs. the official baseline, raising P10–P90 coverage from **68% to 79%** .
- Trained LightGBM quantile and classification models to predict 10-hour price ranges, spike risk and regulation direction.
- Built a trader dashboard and API with plain-language spike alerts, 10-hour P10–P90 price ranges, hindsight replay of past forecasts, on-demand forecasting from uploaded files, and SHAP-based explanations of the top price drivers.

Publications

- **Md Jawadul Hasan**, S. H. Shifat, J. Matubber, R. Hossain, M. A. Rahman, B. M. T. Haque, M. J. Hossen. "An in-depth exploration of machine learning methods for mental health state detection: a systematic review and analysis." *Frontiers in Digital Health*, vol. 7, 2026.
- **Md Jawadul Hasan**, A. Das, J. Matubber, S. H. Shifat, M. K. Morol. "Enhanced Classification of Anxiety, Depression, and Stress Levels: A Comparative Analysis of DASS21 Questionnaire Data Augmentation and Classification Algorithms." *3rd International Conference on Computing Advancements (ICCA 2024)*, ACM.
- M. A. H. Reaj, R. D. Gupta, M. Y. Rahat, N. Fahad, **Md Jawadul Hasan**, T. H. Liew. "GANDiff-FR: Hybrid GAN–Diffusion Synthesis for Causal Bias Attribution in Face Recognition."

Certifications

- Google Data Analytics Professional Certificate – Google (Coursera)

Achievements

- Ranked 6th in the Intra AIUB Programming Contest Fall 2021–22
- Solved 600+ algorithmic challenges on LeetCode, Codeforces, HackerRank, and similar platforms
- Selected for the Dean's List honors for outstanding academic performance
- Finalist in an app development competition hosted by IEEE India in 2020
- Top 1% in class; received Summa Cum Laude for an exceptional CGPA

Languages

English: Fluent

Finnish: Learning, currently enrolled in Finnish for Foreigners 1, Metropolia University of Applied Sciences